

Question number	Answer	Additional Guidance	Mark
	A description that includes the three of the following: • find suitable ages of seedlings (1) • find a suitable temperature for respiration/growth (1) • find a suitable time for measuring a volume of gas (1) • find a suitable {mass / number} of seedlings (to give measurable results)(1) • find a suitable {pH / watering regime} for {respiration / growth} (1)	Ignore practice the method... Accept find a suitable time to give measurable results	(3)

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	An answer that includes eight of the following points: • MP1 clear statement of the dependent variable e.g. volume of gas / distance moved by meniscus (1) • MP2 appropriate description or diagram of apparatus used (1) • MP3 use seedlings of different ages (1) • MP4 record time for a measured distance of the meniscus/volume from gas syringe (1) • MP5 do with and without soda lime (1) • MP6 description of how to find the volume of oxygen used (1) • MP7 temperature needs to be controlled (1) • MP8 use of a thermostatic device to control temperature (1) • MP9 sterilise the (surface of) seeds (1) • MP 10 formula for calculating RQ (1)	Accept RQ Accept test tube with capillary tube / gas syringe the apparatus must be able to work Accept age in days or weeks only Accept record distance meniscus moved in a set time Accept distance x area / reading gas syringe / subtract one gas reading from another to give CO₂ produced Carbon dioxide released ÷ oxygen consumed	(8)

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4c	- table to record data with headings. appropriate units and means calculated from repeats (1) - line/scatter graph format with labelled axes (1) - use of an appropriate (named) statistical test (1)	table should have space or columns for repeats. means can be stated somewhere else. Accept units as appropriate once, either in the table or graph RQ has no unit Accept bar graph if only two ages tested t test if only two ages otherwise a correlation test or named test Allow a stats test that relates to the data collected in the table	(3)

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4d	An answer that includes two of the following: • difficult to measure {distances / volumes of gas} (1) • (difficult to prevent) contamination of mung beans (1) • difficult to decide if seedlings are germinating at the same rate (1) • seedlings may change from aerobic to anaerobic respiration (during investigation) (1)	Ignore amount Accept changes in {atmospheric pressure / temperature} change volume of gas Accept infected Accept seedlings may have a different starch / protein content	(2)

